

Controlled by geasutres glove

Motivation:

Geasture Glove is going to recognize different hand geasures: open hand, fist, pointing, finger combinations, thumb-to-finger touches, wrist tilt/rotation, swipe/flick, shake, and combinations such as `index touch + move hand`.

Glove can be used for different purposes. Mouse cursor control, presentations, control of RC robots, gaming and etc.

Geasture Glove is going to be connected to computer via BLE

"Brain" of the glove is esp32

Limitations:

1. glove is going to had low accuracy in finger bending, because we want to use flex sensors of our own production and theoretically it wont work.

Alternative: we are going to use metal pads on the end of each finger which will conduct conductive tape across the hand to activate exact finger. Than we are going to use this finger combinations as commands combined with IMU value

2. Orientation in space, rotation, movement is not going to be accurate.

Alternative: we are not gonna move hand too much, suggested sensor can detect events like: shake, fast movement in specific direction and etc. But our main sensor is going to be gyroscope, to detect wrist rotation up, down, right and left.

List of required components:

Component	Qty	Purpose	Expected cheap price	Inevitable to buy
ESP32-S3 development board	1	Main MCU, ADC, BLE, USB	€8-12	yes

Component	Qty	Purpose	Expected cheap price	Inevitable to buy
MPU6050 / GY-521	1	Accelerometer + gyroscope	€3-4	yes
Velostat sheet	1 small piece	5 DIY finger flex sensors	€2-5	no
Copper conductive tape	~1 m	Electrodes inside flex sensors	€2-4	yes
Conductive fabric/tape	small piece	Common palm contact	€2-4	yes
Small conductive fingertip pads	5	Finger contact detection	€1-3 total	yes
10 kΩ resistors	5-10	Flex-sensor voltage dividers	€0.20-0.50	yes
10 kΩ / internal pull-ups	0-5	Finger-contact inputs	€0-0.20	yes
100 nF ceramic capacitors	5-10	Sensor noise filtering	€0.20-0.50	yes
3.7 V LiPo 400-800 mAh	1	Battery	€4-7	yes
TP4056 USB-C charger + protection	1	Charging LiPo	€0.50-1.50	yes
Small slide power switch	1	Glove ON/OFF	€0.30-1	yes
JST-PH 2.0 connector	1 pair	Removable battery	€0.40-1	yes
Thin flexible wire, ~28-32 AWG	3-5 m	Finger wiring	€2-4	yes
Cheap fabric glove	1	Base	€2-5	no

Component	Qty	Purpose	Expected cheap price	Inevitable to buy
Heat-shrink tubing	small amount	Protect connections	€1-2	yes
Solder / miscellaneous	—	Assembly	€1-2 worth	yes